



EVENTHUB - SECURE AND SCALABLE EVENT MANAGEMENT SYSTEM USING FULL-STACK TECHNOLOGIES

PHASE-II REPORT

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**in the partial fulfillment of the requirements for the award of degree of
BACHELOR OF TECHNOLOGY**

in

**INFORMATION TECHNOLOGY
RATHINAM TECHNICAL CAMPUS, COIMBATORE
(AUTONOMOUS)**

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APRIL 2026

BONAFIDE CERTIFICATE

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DECLARATION

We, **AIMAN A, AVINASH P, KARTHIK R, and SUSIDHARAN A**, hereby declare that the project report titled **“EVENTHUB - SECURE AND SCALABLE EVENT MANAGEMENT SYSTEM”** done by us under the guidance of **Er. Karthika R M.E., at Rathinam Technical Campus** submitted in partial fulfilment of the requirement for the award of the Bachelor of Technology degree in Information Technology. Certified further that, to the best of our knowledge, the work reported herein does not form part of any other project report or dissertation on the basis Of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ACKNOWLEDGEMENT

The success of work depends on the team and its cooperation. We take this Opportunity to express our gratitude and sincere thanks to everyone who helped us with our project. First and foremost, we would like to thank the Management for the excellent infrastructure, facilities, and constant support for the successful completion of the project work. We wish to express our heartfelt thanks and deep sense of gratitude to our Chairman **Dr. MADAN A SENDHIL M.S., Ph.D.**, for his valuable guidance and continuous support.

My sincere thanks to honorable our **Dr. B. NAGARAJ, M.E., Ph.D.**, Chief Business Officer, Rathinam Group of Institutions for giving us the opportunity to display our professional skills through this project.

Our sincere thanks to honorable Principal **Dr. KRISHNARAJ M.E., Ph.D.**, for allowing us to display our professional skills through this project.

Our special thanks to our **Dr. A RAJAVEL M.E., Ph.D.**, Associate Dean - School of Computing for this valuable guidance and continuous support. and suggestions to improve the quality of the project work.

We are deeply grateful to **Er. R KARTHIKA M.E.**, Head of the Department, Information Technology for his insightful guidance, unwavering support, and for fostering an atmosphere conducive to learning and innovation.

We are greatly thankful to our project Supervisor **Er. R KARTHIKA M.E.**, Assistant Professor, Department of Information Technology, for her dedicated mentorship, continuous encouragement, and the valuable time she devoted to guiding us throughout our study and ensuring the successful completion of our project.

We express our deep sense of gratitude to all the faculty members, and supporting and non-supporting staff members for their continuous support in completing this project work successful.

ABSTRACT

The EVENTHUB project presents a smart, role-based online event management and ticket booking system that addresses the increasing need for secure, structured, and user-friendly digital platforms to handle modern events. It offers administrators a centralized console to create and configure events with rich details such as descriptions, categories, venues, schedules, ticket prices, seating capacity, and visual assets like banners or posters, while end users can seamlessly discover events, view real-time availability, and complete bookings through an intuitive, responsive web interface. The platform is built using a full-stack architecture that combines a dynamic frontend with a Node.js and Express.js backend and a relational database, enabling consistent data validation, transaction-safe storage of users, events, and bookings, and efficient processing of concurrent requests. Security and access control are enforced through JSON Web Token (JWT)-based authentication and role management, ensuring that administrative operations are restricted to authorized organizers and that user accounts and booking data remain protected from unauthorized modifications. Experimental evaluation and functional testing demonstrate that the system can reliably handle common workflows such as sign-up, login, event publishing, ticket booking, and cancellation with low latency and accurate seat tracking, even as the number of events and participants grows. In addition, the modular design of EVENTHUB makes it suitable for future extensions such as online payments, automated notifications, feedback collection, and analytics dashboards, positioning it as a scalable and practical solution for colleges, communities, and organizations seeking to digitize their event operations.

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
CPU	-	Central Processing Unit
RAM	-	Random Access Memory
SSD	-	Solid State Drive
OS	-	Operating System
UI/UX	-	User Interface / User Experience
API	-	Application Programming Interface
DB	-	Database
DBMS	-	Database Management System
JWT	-	JSON Web Token
HTTP	-	HyperText Transfer Protocol
HTTPS	-	HyperText Transfer Protocol Secure
URL	-	Uniform Resource Locator
IDE	-	Integrated Development Environment
MVC	-	Model–View–Controller
REST	-	Representational State Transfer
JSON	-	JavaScript Object Notation
HTML	-	HyperText Markup Language
CSS	-	Cascading Style Sheets
JS	-	JavaScript
SQL	-	Structured Query Language
IP	-	Internet Protocol
CRUD	-	Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

Event management has become an essential activity in educational institutions, organizations, and communities where multiple events are conducted regularly. Traditional methods of managing events, such as manual record keeping or spreadsheet-based tracking, often lead to inefficiencies, data inconsistencies, and difficulties in monitoring participant attendance. These limitations highlight the need for a more organized and automated system that can simplify event planning, registration, and monitoring processes while ensuring accuracy and security.

To address these challenges, the proposed system introduces EVENTHUB, a secure and scalable role-based event management platform. The system enables administrators, organizers, and participants to interact through a unified web interface, allowing efficient event creation, registration, and monitoring. By integrating modern technologies such as QR code and RFID-based entry verification, the system ensures secure participant authentication and accurate attendance tracking. Overall, EVENTHUB aims to improve the efficiency, reliability, and user experience of event management through automation and centralized data management.

The EVENTHUB system is designed to provide a structured and role-based environment where different users can perform their specific tasks efficiently. Administrators have the authority to manage user accounts, monitor events, and maintain overall system operations. Event organizers are responsible for creating and managing event details, handling participant registrations, and ensuring that event activities are properly coordinated. Participants can easily browse available events, register online, and receive confirmation for their participation through the system.

Furthermore, the system focuses on improving the security and transparency of event participation through automated verification and monitoring mechanisms. The integration of QR code and RFID-based identification allows quick validation of participants during event entry, reducing manual verification and minimizing errors. In addition, the centralized database enables real-time data updates, efficient record management, and accurate report generation, making the system reliable and scalable for managing events in educational institutions and organizations.

1.1 Background

Event management plays an important role in organizing activities such as conferences, workshops, cultural programs, technical events, and social gatherings. Traditionally, these events were managed using manual methods like paper-based registrations, physical tickets, and direct communication, which required significant time and effort. As the number and size of events increased, these methods became inefficient and prone to errors such as data loss, duplicate entries, and poor coordination. The growing demand for organized and efficient event handling has created the need for digital solutions.

With the rapid development of web technologies and increased internet usage, event management has gradually shifted toward online platforms. Web-based systems enable organizers to publish event details, manage registrations, and communicate with participants more effectively. At the same time, users benefit from easy access to event information and online booking facilities. However, many existing systems lack proper security, scalability, and user-friendly design, limiting their effectiveness in handling large numbers of users and events.

To address these challenges, EVENTHUB is proposed as a secure and scalable event management system using full-stack technologies. The system provides a centralized platform where administrators can manage events and users can register and book tickets conveniently. By integrating modern frontend and backend technologies with secure authentication mechanisms, EVENTHUB aims to improve efficiency, reduce manual effort, and provide a reliable solution for managing events in a digital environment.

1.2 Problem Statement

Event management has traditionally been handled using manual procedures or fragmented digital tools, which often fail to meet the demands of modern event organization.

Activities such as participant registration, ticket distribution, attendance tracking, and communication are commonly managed through spreadsheets, paper forms, or isolated applications. These approaches consume significant time and effort and are highly susceptible to human errors, including incorrect data entry, duplication of records, and loss of important information. As events grow in size and complexity, managing large volumes of participant data becomes increasingly difficult, resulting in poor coordination and reduced overall efficiency.

Although several online event management platforms exist, many of them lack proper security and structured access control mechanisms. Inadequate separation between administrator and user functionalities can result in unauthorized access to sensitive data such as user details, event schedules, and booking records. The absence of strong authentication and authorization methods increases the risk of data breaches and misuse, which can reduce user confidence and limit the effectiveness of such systems in real-world applications.

Scalability and system adaptability further contribute to the challenges faced by event organizers. Several existing systems are not designed to handle a growing number of users or simultaneous access during peak periods, leading to slow performance, server overload, and system downtime. Additionally, limited customization options and the absence of real-time updates restrict the ability of organizers to modify event details, track bookings efficiently, and respond quickly to changes. These limitations highlight the need for a secure, scalable, and flexible event management platform like EVENTHUB, which can support efficient operations, ensure data security, and adapt to evolving organizational requirements.

1.3 Objectives

1.Event Management Scope

The primary objective of the EVENTHUB project is to design and develop a centralized event management system that simplifies the overall process of organizing events. The system aims to replace traditional manual methods with a digital platform that automates event creation, registration, and ticket booking. By reducing manual effort, the system helps minimize errors and improves operational efficiency. This objective focuses on providing a reliable and well-structured solution for managing events of different types and scales.

Another key objective of the project is to ensure a high level of security for user and event data. The system is designed with secure authentication mechanisms to protect sensitive information from unauthorized access. Role-based access control is implemented to restrict system functionalities based on user roles. This objective helps maintain data confidentiality and integrity throughout the system. Secure handling of login credentials and event records builds trust among users. Overall, the system aims to follow standard web security practices.

2.User Role Management

The project also aims to clearly differentiate the roles and responsibilities of administrators and users. Administrators are provided with exclusive privileges to create, modify, and manage events within the system. Users, on the other hand, are allowed to browse events and book tickets based on availability. This separation of roles helps in maintaining proper control over system operations. It also reduces the risk of accidental or intentional data manipulation. Clear role definition ensures smooth and controlled event management.

Scalability is another important objective of the EVENTHUB system. The platform is designed to handle a growing number of users and events without affecting system performance. By using a modular full-stack architecture, the system can efficiently manage increased data and user requests. This objective ensures that the application remains stable during peak usage periods. It also allows the system to support future growth. Scalability makes EVENTHUB suitable for both small and large events.

3.Data Management and Storage

Enhancing the user experience is a major focus of the project. The system aims to provide a clean, intuitive, and responsive user interface that is easy to navigate. Users should be able to access event details, register, and book tickets without confusion or difficulty. Clear layouts and interactive elements are designed to improve usability. This objective ensures that users have a smooth and satisfying interaction with the platform. A good user experience increases user engagement and adoption.

Another objective of the project is to ensure efficient data storage and management. A structured database is used to store user details, event information, and booking records securely. This approach allows easy data retrieval and modification whenever required. Proper data management helps in maintaining consistency and reliability across the system. It also supports efficient reporting and monitoring. This objective contributes to the overall stability of the application.

4.System Accessibility

The project also aims to demonstrate the effective use of full-stack web development technologies. EVENTHUB integrates frontend interfaces, backend services, and database management into a single system. This objective helps in applying theoretical knowledge to practical development. It also enhances understanding of modern software development practices. The project showcases how different technologies work together to build a complete web application.

Finally, the project aims to develop a flexible and extensible system that can be enhanced in the future. EVENTHUB is designed to support additional features such as online payment integration, notification services, and analytics modules. The system architecture allows easy modification and expansion. This objective ensures that the platform remains relevant as requirements evolve. Future enhancements can be added without major changes to the existing system.

1.4 Scope of the Project

1.Role-Based Access Control

The scope of the EVENTHUB project includes the design and development of a web-based event management system that provides essential functionalities required for organizing and managing events efficiently. The system focuses on digitizing key event-related processes such as event creation, user registration, and ticket booking. It provides a centralized platform where all event information is stored and managed securely. This approach reduces manual effort and improves coordination between event organizers and participants. The project emphasizes reliability and ease of use. It is designed to support various types of events.

The system supports two main user roles: administrators and users. Administrators are responsible for creating events, updating event details, managing ticket availability, and monitoring registrations. Users can register, log in, view available events, and book tickets through the system. Role-based access control ensures that users can access only the functionalities relevant to their role. This clear separation improves system security and operational control. The scope focuses on maintaining a structured and secure workflow. It ensures smooth interaction between different users.

2.Enable Real-Time Event Monitoring

The project also includes the implementation of backend services and database management for efficient data handling. A structured database is used to store user details, event information, and booking records securely. Backend APIs handle authentication, data processing, and communication between the frontend and database. Security mechanisms such as token-based authentication are included within the scope of the project. These features ensure data integrity and controlled access. The system is designed for consistent performance and reliability.

However, the current scope of the project is limited to core event management functionalities. Advanced features such as online payment gateway integration, automated email or SMS notifications, and mobile application support are not included in this phase. These features are identified as future enhancements that can be added later. The project focuses on delivering a functional and scalable prototype suitable for academic evaluation. This defined scope ensures clarity and ease of implementation. It also allows future expansion without major system redesign.

1.5 Significance of the Study

The significance of the EVENTHUB project lies in its ability to address the challenges associated with traditional event management methods. By providing a digital platform for event organization and participation, the system reduces reliance on manual processes such as paper-based registration and physical ticket distribution. This transformation helps minimize human errors and improves overall efficiency. The study highlights how technology can simplify complex organizational tasks. It also demonstrates the importance of automation in modern event management practices.

1.Improved Event Management Efficiency

From an operational perspective, the EVENTHUB system enhances efficiency and coordination for event organizers. The centralized management of event details, user registrations, and bookings allows organizers to monitor and control event activities more effectively. Real-time access to data enables better planning and decision-making. This study emphasizes how a structured event management system can improve workflow management. It also supports better resource utilization during event planning and execution.

The study is significant in terms of data security and system reliability. By implementing secure authentication and role-based access control, the system protects sensitive user and event information. This focus on security increases user trust and ensures data integrity. The project demonstrates the importance of secure design in web-based applications. It highlights how proper security mechanisms can prevent unauthorized access. This aspect is critical in systems handling personal and organizational data.

2.Real-Time Monitoring and Reporting

From an academic standpoint, the EVENTHUB project provides valuable learning experience in full-stack web development. The study integrates frontend design, backend processing, database management, and security features into a single application. It allows students to apply theoretical knowledge to real-world scenarios. This significance lies in enhancing practical understanding of modern software development. It also strengthens problem-solving and system design skills.

Finally, the study contributes to future technological development in the field of event management systems. EVENTHUB is designed with scalability and flexibility in mind, allowing future enhancements such as payment integration, notification services, and analytics features. The project serves as a foundation for further research and development. It demonstrates how scalable systems can adapt to evolving requirements. This significance ensures long-term relevance and practical applicability of the system.

CHAPTER 2

LITERATURE REVIEW

[1] Laudon, K. C., & Laudon, J. P., “Management Information Systems: Managing the Digital Firm,” Pearson Education.

Laudon and Laudon discussed the role of management information systems in improving organizational efficiency through digital transformation. Their work emphasizes how information systems automate administrative tasks such as data handling, scheduling, and user management. In the context of event management, these systems provide a structured approach to organizing events, managing participants, and maintaining records efficiently. The study highlights the importance of centralized platforms in reducing manual workload and improving decision-making. However, the authors note that system effectiveness largely depends on proper security mechanisms and scalable architecture.

[2] Joseph, A., & Shah, S., “Web-Based Event Management Systems: Design and Implementation,” International Journal of Computer Applications, 2020.

Joseph and Shah examined the design and implementation of web-based event management systems aimed at automating event organization processes. Their study focused on how online platforms can simplify event scheduling, participant registration, and information dissemination through centralized web applications. The authors highlighted the effectiveness of role-based access control in separating administrative and user functionalities, which improves system security and operational efficiency.

[3] Smith, R., & Kumar, P., “Full-Stack Ticket Booking Systems: Architecture and Performance Analysis,” Journal of Software Engineering and Applications, 2021.

Smith and Kumar explored the design and development of full-stack ticket booking systems, focusing on both frontend user interfaces and backend service architectures. Their study highlighted the integration of responsive web frameworks with server-side technologies to provide seamless booking experiences for users. Key features analyzed included real-time seat availability, dynamic pricing, payment gateway integration, and notification systems. The authors emphasized the importance of database normalization, caching mechanisms, and RESTful API design to improve system performance and scalability.

[4] Chen, L., & Verma, S., “Secure User Authentication in Web Applications Using JWT,” *International Journal of Information Security*, 2022.

Chen and Verma investigated the implementation of secure authentication mechanisms in web applications using JSON Web Tokens (JWT). Their study focused on how JWT can provide stateless, scalable, and tamper-proof authentication for both single-page and multi-page web applications. The authors highlighted the role of token expiration, signature verification, and refresh tokens in maintaining security while improving user experience. The research demonstrated that JWT-based authentication reduces server-side session management overhead and simplifies API security in distributed architectures. However, they noted potential risks, such as token theft and replay attacks, which require additional safeguards like HTTPS enforcement, secure storage, and token revocation strategies.

[5] Patel, R., & Singh, A., “Design and Development of Database-Driven Web Applications,” *International Journal of Web Engineering*, 2019.

Patel and Singh examined the architecture and implementation of database-driven web applications, emphasizing the interaction between front-end interfaces and back-end databases. Their study focused on how dynamic content generation, CRUD operations, and data persistence can be efficiently managed through relational and NoSQL databases. The authors highlighted the significance of database normalization, query optimization, and secure connection handling to ensure performance, scalability, and data integrity. Their findings indicated that database-driven applications significantly improve user interactivity and administrative efficiency compared to static web pages.

Reference	Focus Area	Key Findings
Laudon & Laudon	Management Information Systems	Improves efficiency and data management
Joseph & Shah (2020)	Web Event Management	Automates scheduling and registration
Smith & Kumar (2021)	Ticket Booking Systems	Real-time booking and scalable design
Chen & Verma (2022)	JWT Authentication	Secure and stateless user login
Patel & Singh (2019)	Database Web Applications	Efficient CRUD operations and scalability

Table 2.1 Summary of literature Review

CHAPTER 3

PROJECT ANALYSIS

3.1 Problem Statement

3.1.1 Existing System

In the existing system, event management is primarily carried out using manual methods or basic digital tools such as spreadsheets, emails, and standalone registration forms. These approaches require significant human involvement for tasks such as event scheduling, participant registration, and ticket distribution. As a result, the process becomes time-consuming and prone to errors such as duplicate entries, incorrect data handling, and loss of important information. Managing large-scale events using such methods often leads to confusion and inefficient coordination among organizers.

Additionally, existing systems lack centralized control and real-time data updates. Event details and participant information are often scattered across multiple platforms, making monitoring and management difficult. The absence of proper role separation between administrators and users increases the risk of unauthorized access and data misuse. Due to limited security features and poor scalability, existing systems are unable to efficiently handle increasing user demand and multiple concurrent events.

3.1.2 Challenges of the Existing System

One of the major challenges of the existing event management systems is their heavy reliance on manual processes and disconnected digital tools. Activities such as event registration, ticket allocation, and participant tracking are often handled using spreadsheets, paper forms, or separate online platforms. This lack of integration leads to data inconsistency, duplication of records, and difficulty in maintaining accurate event information. Manual intervention at multiple stages increases the possibility of errors and delays, especially when managing large-scale events with a high number of participants.

Another significant challenge is the lack of security, scalability, and real-time management capabilities in existing systems. Many systems do not implement proper role-based access control, allowing unauthorized users to access or modify sensitive event data. In addition, these systems are not designed to handle a growing number of users, resulting in slow performance and system failures during peak usage. The absence of real-time updates makes it difficult for organizers to monitor registrations, manage seat availability, and respond quickly to changes. These challenges highlight the need for a secure, scalable, and centralized event management solution like EVENTHUB.

Overall, these challenges reduce the efficiency and reliability of traditional event management approaches. Without proper security, scalability, and centralized control, existing systems fail to meet the demands of modern event organization. This creates a strong need for an improved and automated solution.

3.2 Proposed Solution

3.2.1 Introduction to the Smart Vision–Driven Occupancy Detection System

The proposed system, EVENTHUB, aims to overcome the limitations of the existing system by providing a secure and centralized web-based event management platform. The system automates essential processes such as event creation, user registration, and ticket booking through a single interface. By integrating frontend and backend technologies, EVENTHUB ensures smooth data flow and real-time updates. This approach significantly reduces manual effort and improves accuracy in event management operations. The modular architecture allows efficient handling of multiple user requests and real-time data processing. EVENTHUB also ensures consistent system reliability by maintaining structured data flow between the frontend, backend, and database.

Furthermore, the proposed system implements role-based access control and secure authentication mechanisms to ensure the protection of sensitive user and event data. Administrators are granted exclusive privileges to create, update, and manage event details, monitor registrations, and control ticket availability, while general users are restricted to viewing events and booking tickets based on availability. This clear separation of roles helps prevent unauthorized access and reduces the risk of data manipulation. Secure authentication mechanisms are incorporated to validate user identities and safeguard login credentials.

3.2.2 Core Functionalities of the Occupancy Detection System

The EVENTHUB system offers essential functionalities that automate and simplify the process of event management. Administrators are provided with features to create, update, and manage events by defining details such as event name, schedule, location, description, and ticket availability. The system allows efficient handling of multiple events through a centralized platform, ensuring that all event-related information is accurate and up to date. This functionality reduces manual workload and improves overall coordination in event organization.

EVENTHUB also supports secure user registration, authentication, and ticket booking functionalities. Users can register, log in securely, browse available events, and book tickets based on availability. Role-based access control ensures that administrative features are restricted to authorized users only. All data related to users, events, and bookings are stored and managed securely with real-time updates. These core functionalities make EVENTHUB a reliable, user-friendly, and scalable event management solution.

3.2.3 Advantages of EVENTHUB Over Existing Event Management platform

Feature	Existing System	Proposed System
Event Monitoring	Manual tracking using spreadsheets and emails	Centralized, real-time event monitoring
Registration & Booking	Manual tracking using spreadsheets and emails	Automated Registration and Ticket Booking
Scalability	Difficult to manage large-scale events	Scalable system supporting multiple events and users
Security	Limited or no role-based access control	Secure authentication with role-based access

Table 3.1 Advantages of EVENTHUB Over Existing Event Management Platform

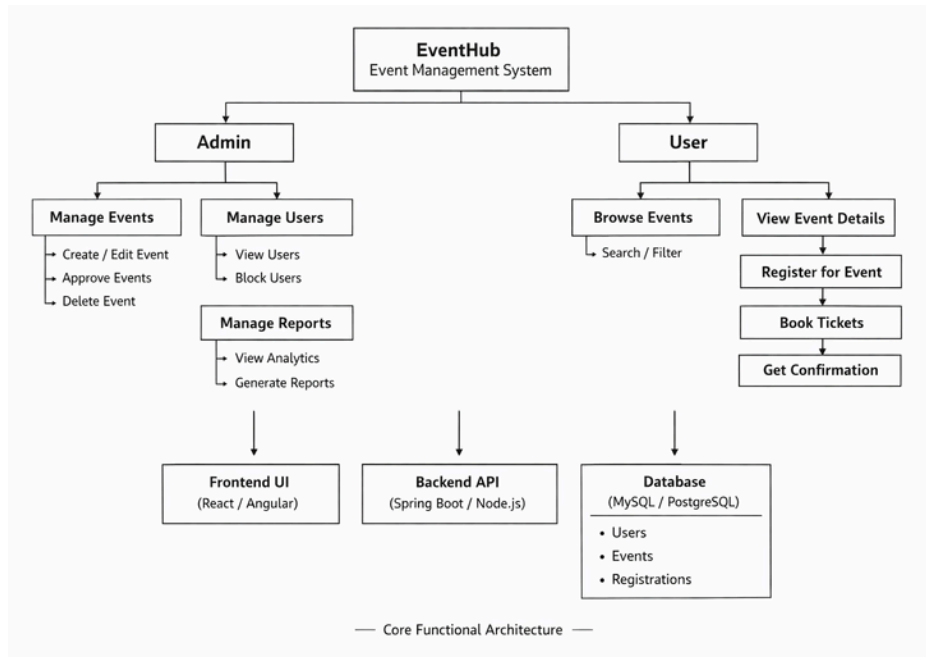


Fig 3.1 Core Functionalities

1. User Management

The system allows users to register and create an account using their personal details such as name, email, and password. After registration, users can securely log in to the platform to access available events. Users can also manage their profile information and view their event registration history. Authentication and authorization mechanisms ensure that only authorized users can access specific features of the platform.

2. Event Registration and Ticket Booking

Users can register for events directly through the platform. Once a user selects an event, they can complete the registration process and book tickets. The system records the registration details in the database and updates the available event capacity to prevent overbooking.

3. Event Browsing and Search

Users can browse all available events listed in the system. The platform provides options to search and filter events based on parameters such as event name, category, date, or location. This helps users quickly find events that match their interests.

3.2.4 Key Benefits of Implementing the Smart Vision–Driven System

One of the most significant benefits of implementing the EVENTHUB system is the improvement in efficiency and accuracy of event management processes. The system automates essential tasks such as event creation, participant registration, and ticket booking, which traditionally require extensive manual effort and coordination. By eliminating manual data entry and paperwork, the system reduces the chances of errors such as duplicate registrations, incorrect participant details, and missing records. EVENTHUB provides a centralized platform that allows organizers to manage multiple events simultaneously and monitor registrations in real time. This automation improves workflow efficiency and ensures better coordination among event organizers. As a result, events can be planned and executed in a more structured and reliable manner.

Another key benefit of the EVENTHUB system is enhanced security and protection of sensitive data. The system incorporates secure authentication mechanisms along with role-based access control to ensure that only authorized users can access specific functionalities. Administrative operations such as event modification and registration management are restricted to authorized personnel, reducing the risk of unauthorized access and data misuse. Sensitive information such as user credentials, event details, and booking records is stored securely and handled in a controlled environment. This strong focus on security increases system reliability and builds trust among users. By following standard web security practices, EVENTHUB ensures data confidentiality, integrity, and availability throughout the event management process.

The proposed EVENTHUB provides an efficient and automated platform for managing events through a centralized web-based system. It simplifies the process of event creation, registration, and participation by allowing administrators and organizers to manage events while participants can easily browse and register for available events. The implementation of role-based access control ensures that each user type has specific permissions, improving system security and maintaining proper control over event operations.

Scalability and improved user experience are also major benefits of implementing the EVENTHUB system. The platform is designed to handle an increasing number of users and events without affecting performance or system stability. Its modular architecture allows efficient processing of multiple user requests and real-time data updates. The user-friendly interface enables participants to easily browse events, register, and book tickets without complexity. At the same time, administrators can efficiently manage large-scale events using structured dashboards and real-time monitoring tools. These features make EVENTHUB a flexible, scalable, and user-centric solution suitable for modern digital event management requirements.

In addition, the system enhances event security and attendance tracking through the integration of QR code and RFID-based entry verification. This hybrid verification mechanism enables quick and reliable participant validation during event entry. The centralized database stores user information, event details, and attendance records, allowing real-time monitoring and report generation. Overall, the system reduces manual effort, minimizes human errors, and provides a scalable and user-friendly solution for modern event management.

CHAPTER 4

SYSTEM SPECIFICATION

4.1 Hardware Specification

To support the efficient operation of the EVENTHUB system, the hardware requirements must be capable of handling web application processing, database operations, and concurrent user access. The system requires a reliable processor to manage backend services, a sufficient amount of memory for smooth multitasking, and fast storage to ensure quick data retrieval. Stable network connectivity is essential to support real-time interaction between users and the server. This hardware setup ensures consistent system performance and reliability during peak usage.

Component	Specification
Processor	Intel Core i5 / i7 (Quad-Core)
RAM	16GB DDR4
Storage	256GB / 512GB SSD
Camera	Web Application Server
Network	High-speed LAN / Wi-Fi for streaming

Table 4.1 Hardware Specification

4.2 Software Specification

The EVENTHUB system is developed using modern full-stack web technologies to ensure reliability, security, and scalability. The frontend of the application is built using HTML, CSS, and JavaScript to provide an interactive and user-friendly interface. Backend services are implemented using Node.js and Express.js to handle business logic and API requests efficiently. A relational database management system is used to store and manage user, event, and booking data securely. Authentication is implemented using JSON Web Tokens (JWT) to ensure secure access control.

4.2.1 Development Environment

The development environment for the EVENTHUB system is designed to support efficient full-stack application development and testing. The project is developed using widely adopted integrated development environments that provide features such as code editing, debugging, and version control support. Frontend development is carried out using standard web technologies, including HTML, CSS, and JavaScript, which enable the creation of responsive and interactive user interfaces. Backend development is implemented using Node.js and Express.js to handle server-side logic and API communication. A relational database management system is used to store and manage application data securely. This environment ensures smooth integration between frontend, backend, and database components.

The system is developed and tested on commonly used operating systems to ensure compatibility and stability. Development tools such as package managers and testing utilities are used to manage dependencies and validate application functionality. Local servers are utilized during development to test application workflows and performance. This setup allows developers to identify and resolve issues efficiently. The development environment supports scalability and future enhancements. Overall, it provides a reliable platform for building and maintaining the EVENTHUB system.

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Node.js
Database	PostgreSQL
Authentication	JSON Web Token (JWT)

Table 4.2 Development Environment

4.3 System Architecture

The system architecture of EVENTHUB follows a three-tier structure comprising the presentation layer, application layer, and database layer for better scalability and maintainability. The presentation layer is built using HTML, CSS, and JavaScript to provide user interfaces for administrators and users to manage and book events. The application layer uses Node.js and Express.js to handle business logic, RESTful APIs, authentication, and role-based access control.

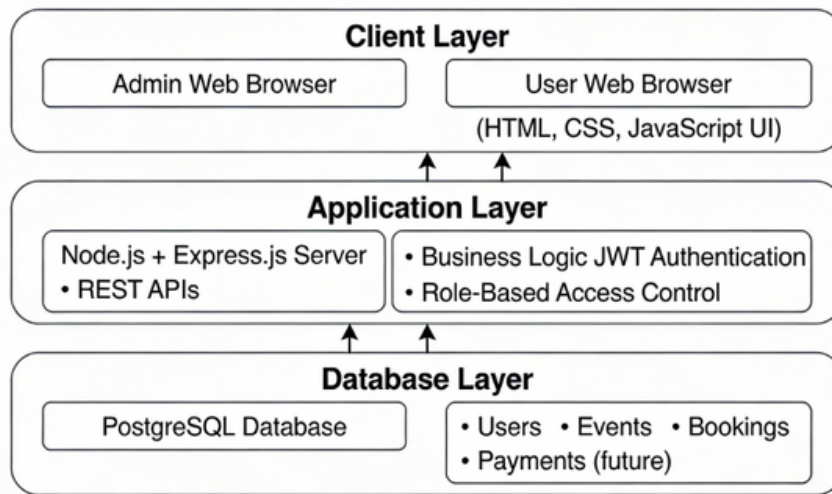


Fig 4.1 System Architecture

4.4 Security and Compliance

The EVENTHUB system incorporates multiple security mechanisms to protect user data, event information, and booking records from unauthorized access and misuse. Authentication is implemented using JSON Web Tokens (JWT), ensuring that only verified users and administrators can access protected APIs and perform sensitive operations such as event creation, modification, and booking management. Role-based access control (RBAC) is enforced at the application layer to clearly separate admin and user privileges, preventing normal users from accessing administrative functionalities. All communication between the client and server is performed over HTTP or can be upgraded to HTTPS in deployment to safeguard credentials and session tokens from interception.

Additionally, input validation, password hashing, and secure database queries are used to reduce common web vulnerabilities such as SQL injection and credential theft, thereby improving overall system reliability and compliance with standard web security practices.

4.5 Integration with External Services

4.5.1 Payment Gateway & Data Storage Integration

To extend EVENTHUB beyond a basic event listing platform, the system can be integrated with external payment gateway services such as Razorpay, Stripe, or similar providers to handle secure online ticket payments. During the booking process, the application layer communicates with the payment gateway through REST APIs, sending details such as user information, event ID, ticket amount, and transaction reference. On successful payment, the gateway returns a confirmation response, which is stored in the EVENTHUB database along with booking details, enabling organizers to track paid bookings and generate basic reports.

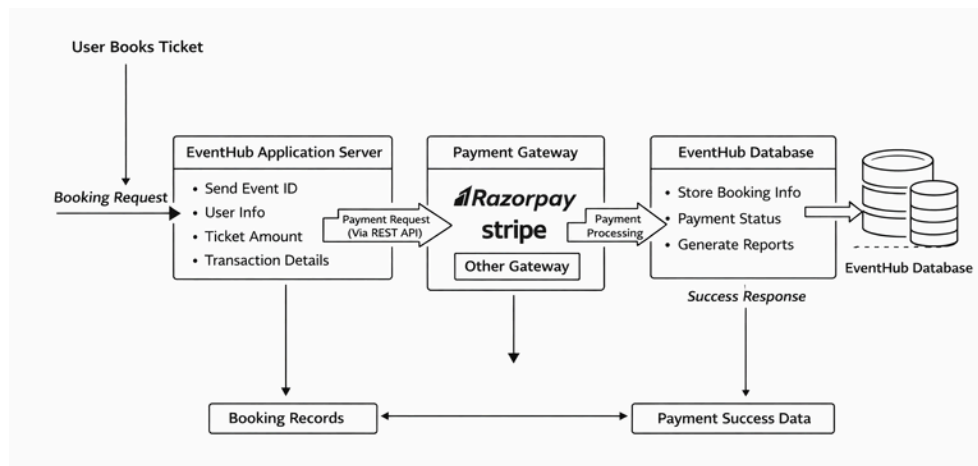


Fig 4.2 Payment Architecture

4.5.2 Deployment Options

The EVENTHUB system can be deployed in multiple environments to ensure flexibility, scalability, and ease of maintenance. During development and testing, the application runs on a local machine using Node.js for the backend and a local PostgreSQL instance, allowing developers to verify features and fix issues quickly. For production, EVENTHUB can be hosted on a cloud platform such as AWS or similar providers, where the Node.js server runs on a virtual machine or managed service and the database is hosted using a managed relational database service. Containerization tools like Docker can be used to package the application and its dependencies into portable images, ensuring consistent behavior across different machines and simplifying deployment.

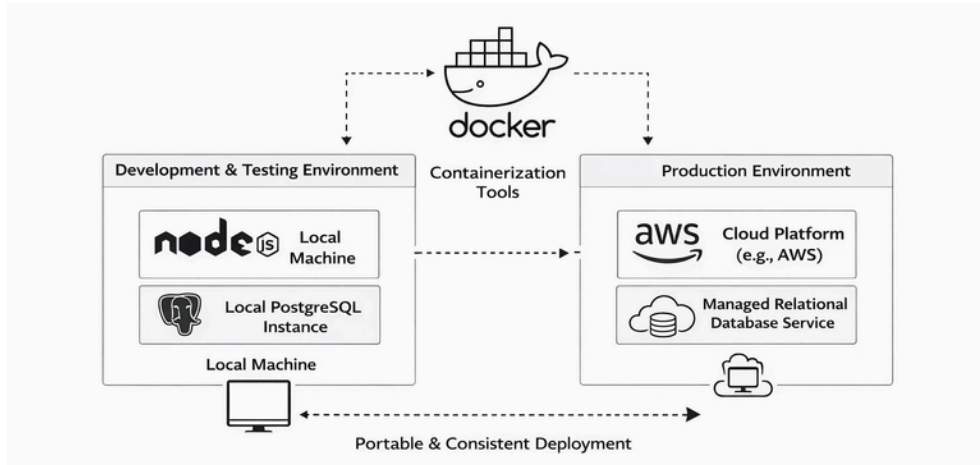


Fig 4.3 Deployment Architecture

4.6 Conclusion

This chapter presented the system specification of EVENTHUB, including hardware requirements, software stack, architecture, security mechanisms, and external service integrations. The defined hardware and software specifications ensure that the application can handle web requests, database operations, and concurrent user interactions reliably. The three-tier architecture, combined with JWT-based authentication and role-based access control, provides a secure and scalable foundation for event management operations.

CHAPTER 5

PROJECT DESCRIPTION

5.1 Existing system and limitation

In the existing scenario, event management is commonly handled using manual processes or simple digital tools like spreadsheets, shared documents, and standalone online forms, which require significant human effort for data entry, consolidation, and tracking. Event organizers maintain participant details, ticket availability, and payment status separately, making it difficult to maintain accurate, up-to-date information and increasing the chances of errors such as duplicate registrations or missing records. Communication with participants is often done through disconnected channels such as email, messaging apps, or social media posts, leading to inconsistent information, delayed updates, and confusion during event changes or cancellations.

Furthermore, the absence of strong authentication, authorization, and proper role separation between administrators and users limits data security and makes it difficult for existing setups to scale efficiently as the number of events and participants grows.

5.1.1 Manual Inspection

Manual inspection is one of the most commonly used methods in traditional event management systems, where organizers or volunteers physically verify registrations and tickets at entry points.

Key features:

- Simple to implement and does not require advanced technical infrastructure.
- Staff can directly interact with participants and handle on-the-spot clarifications or issues.
- Immediate human presence allows basic crowd guidance and queue management.

Limitations

- Highly error-prone, subjective, and inconsistent, especially during large events.
- Cannot be performed efficiently for all participants at peak times.
- Difficult to maintain accurate, real-time records of attendees, which affects reporting.

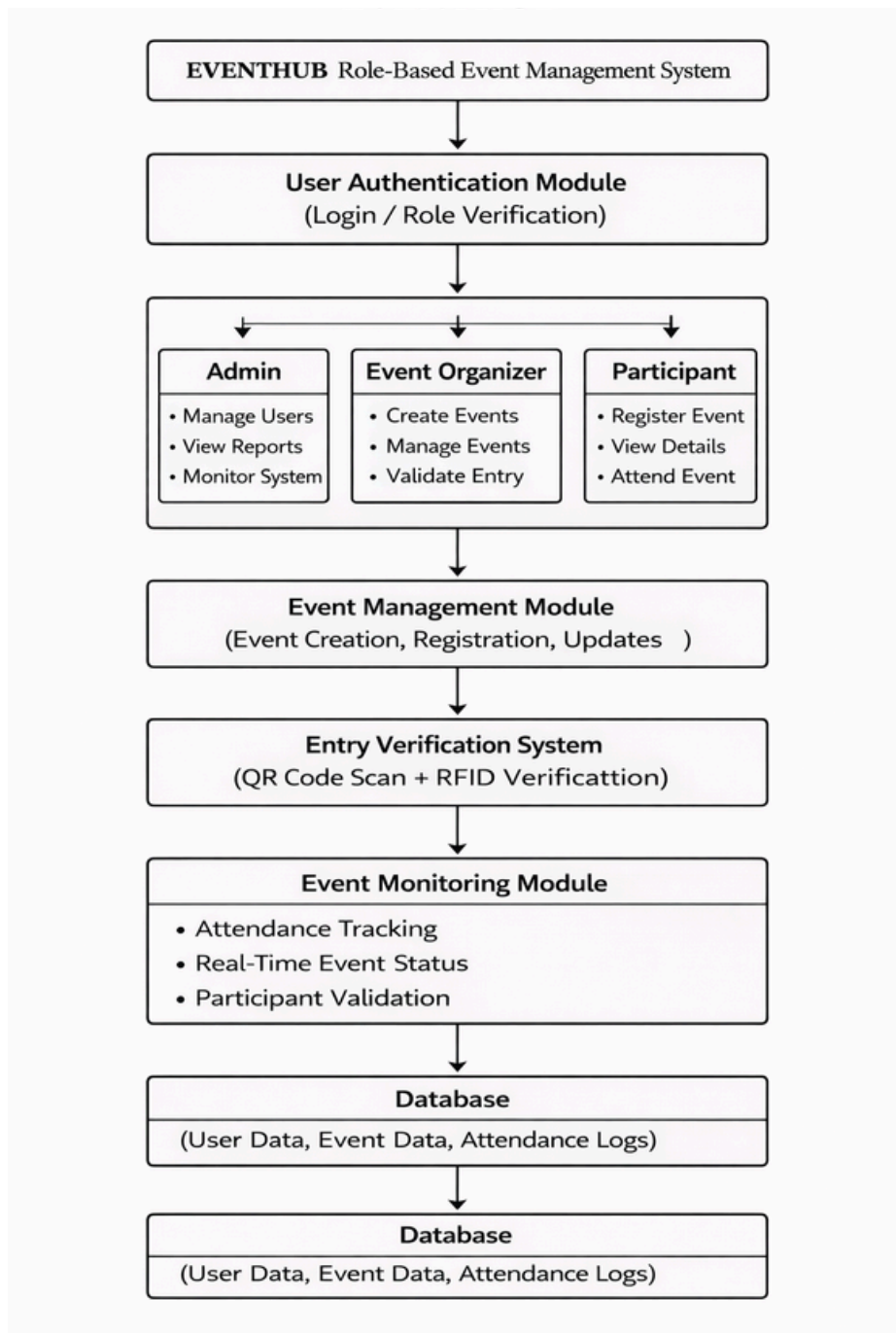


Fig 5.1 Portal Description

5.1.2 Spreadsheet - based management

In many existing setups, events are managed using generic tools such as spreadsheets (Excel, Google Sheets) and shared documents to record participant details, payments, and attendance.

Key features:

- Simple to start with, using widely available office tools without additional software.
- Easy to create custom columns for different event types and export basic reports.
- Supports basic filtering and sorting to search for participants or bookings.

Limitations:

- High risk of data inconsistency when multiple versions of the same sheet are shared. No built-in authentication or role control, so files can be accidentally modified or deleted.
- Does not provide real-time ticket availability, automated validation.

5.1.3 Email - based registration

Many events rely on email-based registration, where participants send their details to a designated email address and organizers manually confirm and record each registration.

Key features:

- Easy to set up using existing email accounts without additional tools.
- Allows participants to share additional queries or special requests along with registration.
- Useful for limited doorway-based monitoring.

Limitations:

- Difficult to track and organize large volumes of emails, leading to missed or delayed.
- No automatic validation, ticket generation, or real-time booking status for participants.
- Prone to miscommunication, duplication, and loss of data.

5.1.4 WhatsApp / Social Media–Based Coordination

Some events use WhatsApp groups or social media messages to collect registrations, where participants send their details through chat and organizers manually maintain separate records.

Key features:

- Enables fast communication and quick sharing of event updates, posters, and reminders.
- No structured database, ticket generation, or automatic limit on registrations.
- Useful for small-scale events or closed communities such as college clubs.

Limitations:

- Difficult to search and track confirmed registrations inside long chat histories.
- No structured database, ticket generation, or automatic limit on registrations.
- Prone to missed messages, duplicate entries, and confusion.

5.2 Proposed methodology

The proposed EVENTHUB system introduces a streamlined, web-based methodology designed to replace outdated and inefficient event management practices with a centralized digital platform. Instead of depending on manual registration, paper forms, or scattered spreadsheets, the system uses structured web forms, secure authentication, and automated booking workflows to handle event-related operations end to end. Its methodology combines real-time data updates, role-based access control, and a responsive user interface so that both organizers and participants can interact with the platform efficiently and reliably. Together, these components ensure that event organizers maintain continuous visibility over registrations, capacity, and basic booking status, reducing coordination issues and improving overall event experience

At the core of this methodology is a full-stack architecture built using HTML, CSS, and JavaScript on the frontend, with Node.js and Express.js managing the backend services. The backend exposes RESTful APIs for functionalities such as user registration, login, event creation, event listing, and ticket booking, while JSON-based communication ensures lightweight and consistent data exchange between client and server. Authentication and authorization are implemented using JSON Web Tokens (JWT), allowing the system to validate users securely and enforce separate privileges for administrators and normal users. This design keeps the application responsive and scalable, making it suitable for environments where the number of events and users may grow over time, such as college festivals or institutional programs.

Complementing the core logic is a relational database (such as PostgreSQL) that stores users, events, and booking records in well-structured tables, ensuring data integrity and easy retrieval. Each booking operation updates event capacity and generates a persistent record, enabling organizers to review participation levels, manage cancellations, and prevent overbooking. Additional modules, such as image upload for event posters and optional integration with payment gateways or email notifications, can be layered on top of this foundation without changing the core workflow.

5.2.1 Admin - side event management

The admin-side module forms the core of EVENTHUB by enabling authorized organizers to configure and control all event-related data through a secure web interface. Instead of manually maintaining separate files or forms, administrators log in to the system and use structured screens to create events by specifying details such as event name, category, description, date and time, venue, ticket price, and seating capacity.

1. Event Creation & Configuration

In this step, the admin fills out a standardized event form where all mandatory fields are validated before submission, ensuring that incomplete or incorrect details are not stored. Images or posters can optionally be uploaded to improve event visibility on the user side, and capacity fields are used later to control booking limits. Once submitted, the event data is stored in the database and immediately becomes available to authenticated users for browsing and booking.

2. Event Update & Status Control

Admins can edit existing events to modify timing, venue, description, or ticket price when changes occur, with updates reflected instantly in the user interface. They can also enable or disable bookings by changing the event status (e.g., “Open,” “Closed,” or “Cancelled”), which helps prevent overbooking and keeps users informed about availability.

3. Booking Overview & Basic Analytics

A dedicated dashboard allows admins to view the list of participants for each event, along with booking count, remaining seats, and basic summary information.

Key Advantages:

- Provides centralized event data, making monitoring makes management easier.
- Reduce manual error by automating registration, booking and basic validation.
- Improves user experience through a consistent web interface for browsing events.

5.2.2 Real time - event tracking & Alerts

The event-tracking module continuously monitors key activities such as new registrations, ticket bookings, cancellations, and capacity usage for each event in EVENTHUB. Whenever a user books or cancels a ticket, the system instantly updates the remaining seat count in the database and reflects the change on both the admin dashboard and user interface

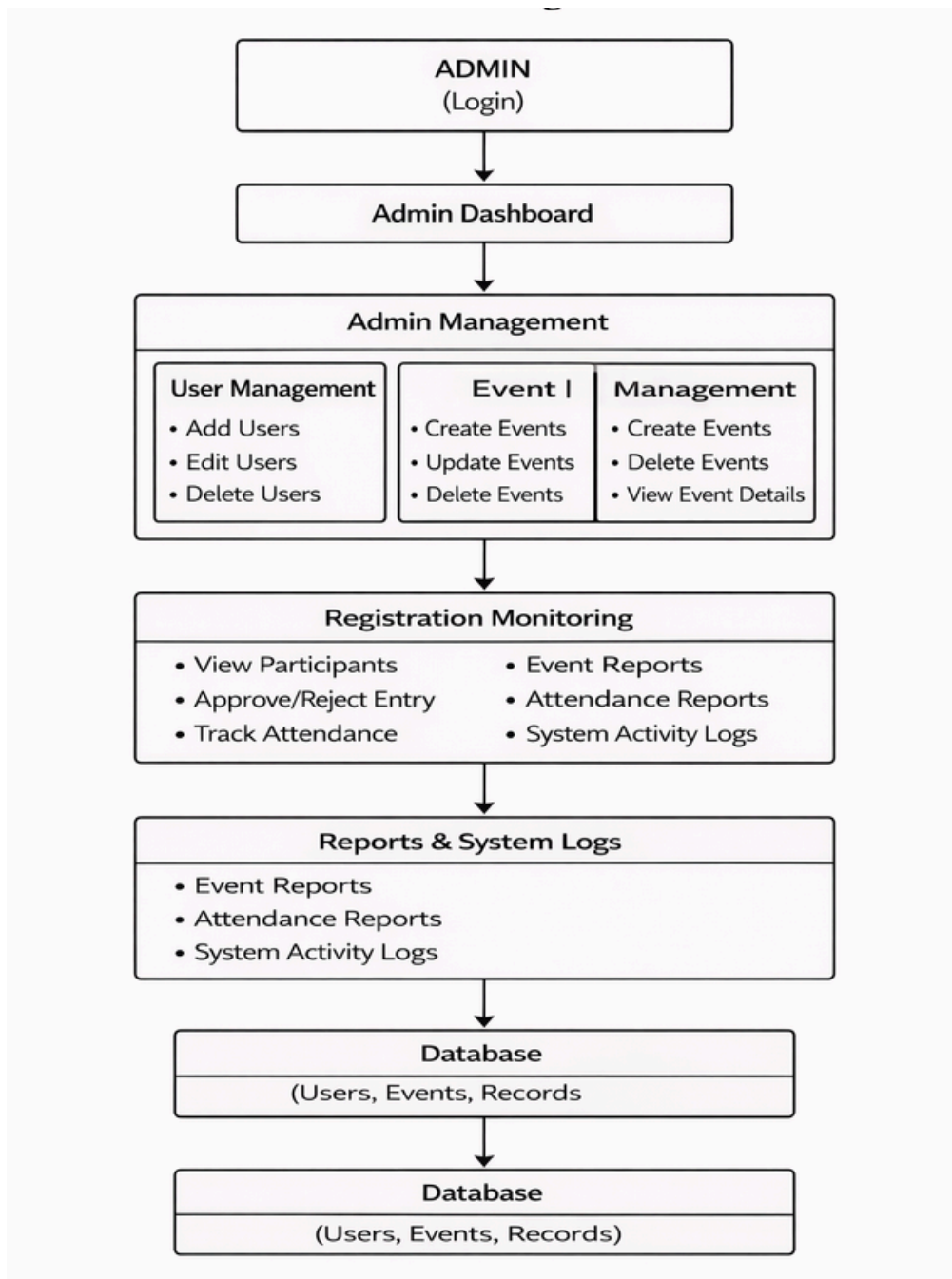


Fig 5.2 Admin Description

1. Live Update Processing

Each booking or cancellation request is processed immediately through the backend APIs, ensuring that event statistics such as “seats filled” and “seats remaining” stay accurate in real time

2. Threshold-Based Notifications

If bookings reach predefined thresholds (for example 80% or 100% capacity), the system can trigger visual alerts on the admin dashboard, such as “Nearly Full” or “Sold Out,” helping organizers react quickly.

3. Dashboard Presentation

The admin dashboard clearly displays live metrics including total registrations, available seats, and booking trends per event, enabling quick monitoring during peak registration periods

Benefits:

- Enables early detection of overbooking or low-registration events.
- Helps organizers manage event capacity and logistics more effectively.
- Reduces manual tracking effort and improves reliability of real-time event information.

5.2.3 System Compatibility & Module Integration

The EVENTHUB system is designed to be modular, flexible, and easily deployable across different institutional and organizational environments, ranging from small college events to multi-day conferences. Its architecture supports both local deployment for development/testing and cloud-based deployment for real-time public access, ensuring that the same codebase can be reused across scenarios.

Integration Features

1. Frontend Compatibility

EVENTHUB works seamlessly on modern web browsers across desktops, laptops, and mobile devices using responsive HTML, CSS, and JavaScript interfaces. This ensures that both organizers and participants can access the platform without installing additional software.

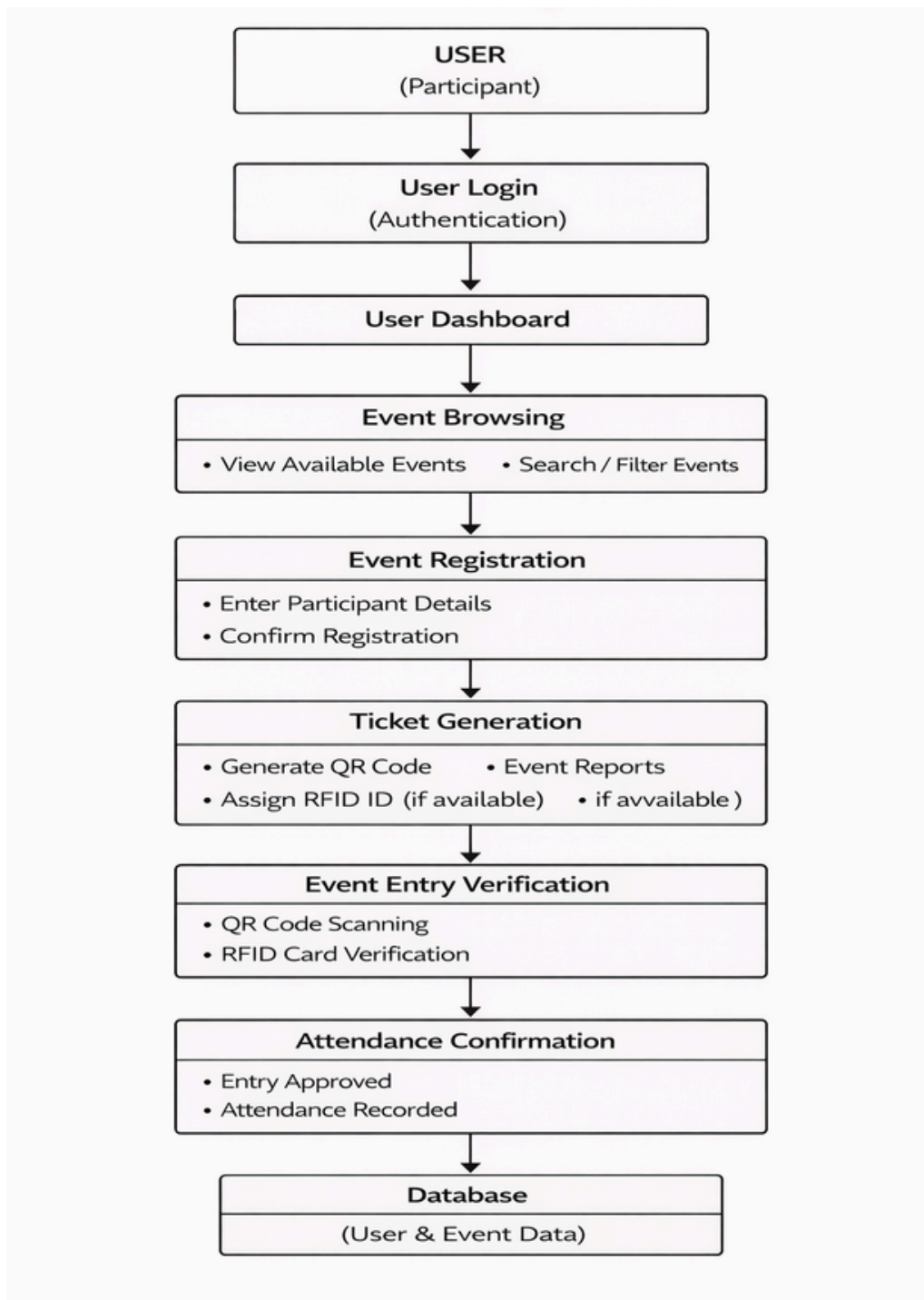


Fig 5.3 User Description

2.Backend–Database Synchronization

The Node.js–based backend handles API routing and synchronization between business logic and the PostgreSQL database, ensuring that any change in events, users, or bookings is instantly reflected in the stored data and user views. The modular nature of the system allows easy integration of additional features such as payment gateway integration, email/SMS notifications, feedback collection, and analytics dashboards.

Advantages:

- Can scale from a single departmental event to institution-wide event management.
- Requires minimal client-side setup since it runs entirely in the browser.
- Can be extended and upgraded with advanced modules without major redesign of the core system.

5.2.4 Data flow & Processing pipeline

The EVENTHUB system follows a simple and efficient data flow pipeline that connects the frontend, backend, and database to provide consistent event and booking information. When a user or admin performs an action—such as creating an event, registering, or booking a ticket—the browser sends a request to the Node.js backend via RESTful APIs. The backend validates the input, applies business rules (like capacity checks or role permissions), and then performs the corresponding database operations on tables such as users, events, and bookings

The data flow in the EVENTHUB system begins when users interact with the web interface to perform actions such as logging in, registering for events, or managing event details. User requests are sent to the application server, where the system processes the inputs based on the user's role and permissions. Administrators can manage users and events, organizers can create and update event information, and participants can browse and register for available events. All user interactions and event-related information are securely transmitted and stored in the centralized database.

5.2.5 Dashboard & Visualization System

The admin dashboard in EVENTHUB provides a clear, web-based interface for monitoring key event metrics such as total registrations, remaining seats, and event status in near real time. Built using HTML, CSS, and JavaScript, it presents tabular views of events and bookings along with action controls for editing, closing, or cancelling events. Visual indicators, such as status labels (“Open,” “Closed,” “Nearly Full”), help organizers quickly understand the state of each event and take timely decisions.

The dashboard and visualization module provides a centralized interface that allows administrators and organizers to monitor and manage event activities efficiently. The dashboard displays key information such as the total number of registered users, active events, participant registrations, and attendance records. Through an intuitive layout, users can quickly access important system functions, view event status, and manage event-related tasks without navigating through multiple pages.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully implements a secure and scalable web-based event management system that digitizes core processes such as event creation, user registration, and ticket booking through a centralized platform. By integrating a full-stack architecture using HTML, CSS, JavaScript, Node.js, Express.js, and a relational database, the system overcomes limitations of traditional manual and spreadsheet-based methods, reducing human errors and improving coordination between organizers and participants. Role-based authentication with JWT and structured database management ensure that user and event data are handled securely while maintaining reliable performance for multiple events and users. The developed prototype demonstrates how modern web technologies can streamline event organization, enhance user experience, and provide a strong foundation for further extensions such as payments, notifications, and analytics.

6.2 Future Work

Future enhancements for the EVENTHUB system focus on improving usability, expanding automation, and enabling wider deployment across institutions and organizations. Advanced analytics modules can be added to generate reports on event attendance, popular categories, and user engagement, helping organizers make data-driven decisions for future planning. Integration with online payment gateways and automated email/SMS notifications will further streamline the booking process by enabling instant payment confirmation, reminders, and digital tickets.

6.3 Final Remarks

The EVENTHUB project demonstrates how a well-designed full-stack web application can modernize event management by replacing fragmented manual processes with a unified, automated platform. Through its secure authentication, structured workflows, and real-time updates, the system improves reliability for organizers and convenience for participants in academic and organizational settings. Beyond fulfilling academic requirements, the project provides practical exposure to end-to-end software development, from requirements analysis and architecture design to implementation and testing, thereby strengthening technical and problem-solving skills

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APPENDIX I

SOURCE CODE

1.BOOKING.JS

```
const express = require('express');
const pool = require('../db');
const {authenticateToken}= require('../middleware/authmiddleware');
const { body, validationResult } = require('express-validator');
const router = express.Router();
// --- 1. LOGGING MIDDLEWARE ---
// This will print the request body to your console for every request to this route
const logRequestBody = (req, res, next) => {
  console.log(`\n[${new Date().toISOString()}] 📧 Incoming ${req.method} request to
${req.originalUrl}`);
  console.log('📦 Request Body:', JSON.stringify(req.body, null, 2));
  next();
};
// GET ORGANIZER BOOKINGS (Bookings for events hosted by the logged-in organizer)
router.get('/organizer', authenticateToken, async (req, res) => {
  try {
    const organizerId = req.user.userId || req.user.id;
    const result = await pool.query(
      `SELECT b.*, e.title as event_title, e.event_date, u.name as user_name, u.email as
user_email
      FROM bookings b
      JOIN events e ON b.event_id = e.id
      JOIN users u ON b.user_id = u.id`
    );
  } catch (error) {
    console.error('Error in GET ORGANIZER BOOKINGS:', error);
    res.status(500).json({ message: 'Internal server error' });
  }
});
```

```

    WHERE e.organizer_id = $1
    ORDER BY b.created_at DESC`,
    [organizerId]);
res.json({ bookings: result.rows });
} catch (error) {
    console.error('Error fetching organizer bookings:', error);
    res.status(500).json({ error: 'Failed to fetch bookings' });}
});

// GET USER BOOKINGS
router.get('/', authenticateToken, async (req, res) => {
    try {
        const userId = req.user.userId || req.user.id;
        const result = await pool.query(
            `SELECT b.*, e.title, e.event_date, e.location, e.image_url
            FROM bookings b
            LEFT JOIN events e ON b.event_id = e.id
            WHERE b.user_id = $1
            ORDER BY b.created_at DESC`,
            [userId]
        );
        res.json({ bookings: result.rows });
    } catch (error) {
        res.status(500).json({ error: 'Failed to fetch bookings' });
    }
});

// CANCEL BOOKING
router.put('/:id/cancel', authenticateToken, async (req, res) => {
    const client = await pool.connect();

```

```

try {
  await client.query('BEGIN');
  const { id } = req.params;
  const userId = req.user.userId || req.user.id;
  const bookingResult = await client.query(
    'SELECT * FROM bookings WHERE id = $1 AND user_id = $2',
    [id, userId]
  );
  if (bookingResult.rows.length === 0) {
    await client.query('ROLLBACK');
    return res.status(404).json({ error: 'Booking not found' });
  }
  const booking = bookingResult.rows[0];
  if (booking.booking_status === 'cancelled') {
    await client.query('ROLLBACK');
    return res.status(400).json({ error: 'Already cancelled' });
  }
  await client.query(
    "UPDATE bookings SET booking_status = 'cancelled' WHERE id = $1",
    [id]
  );
  await client.query(
    'UPDATE events SET available_seats = available_seats + $1 WHERE id = $2',
    [booking.seats_booked, booking.event_id]
  );

  await client.query('COMMIT');
  res.json({ message: 'Booking cancelled successfully' });
} catch (error) {

```

```

await client.query('ROLLBACK');
res.status(500).json({ error: 'Cancellation failed' });
} finally {
  client.release();});});
// --- 2. VALIDATION RULES ---
const validateBooking = [
  body('user_id')
    .notEmpty().withMessage('user_id is missing')
    .isInt().withMessage('user_id must be an integer'),
  body('event_id')
    .notEmpty().withMessage('event_id is missing')
    .isInt().withMessage('event_id must be an integer'),
  body('number_of_seats')
    .notEmpty().withMessage('number_of_seats is missing')
    .isInt({ min: 1 }).withMessage('number_of_seats must be at least 1'),
  body('total_price')
    .notEmpty().withMessage('total_price is missing')
    .isFloat({ min: 0 }).withMessage('total_price must be a positive number'),
  body('ticket_holder_name')
    .notEmpty().withMessage('ticket_holder_name is missing')
    .isString().withMessage('ticket_holder_name must be a string'),
  body('ticket_holder_email')
    .notEmpty().withMessage('ticket_holder_email is missing')
    .isEmail().withMessage('ticket_holder_email must be a valid email'),
  body('ticket_holder_mobile')
    .notEmpty().withMessage('ticket_holder_mobile is missing'),
  // Payment fields (required for your logic, even if not in bookings table)
  body('payment_method').notEmpty().withMessage('payment_method is missing'),

```

```
// --- 3. Apply Middleware and Validation ---

router.post('/', logRequestBody, authenticateToken, validateBooking, async (req, res) => {

  // Check for validation errors

  const errors = validationResult(req);

  if (!errors.isEmpty()) {

    console.error('✗ Validation Failed:', errors.array());

    return res.status(400).json({

      error: 'Missing required fields',

      details: errors.array() // This sends the specific missing field back to frontend

    });

  }

  const client = await pool.connect();

  try {

    await client.query('BEGIN');

    const {

      user_id, event_id, number_of_seats, total_price, status,

      ticket_holder_name, ticket_holder_email, ticket_holder_mobile,

      payment_method, transaction_id, amount

    } = req.body;

    // 1. Insert into bookings table

    const bookingQuery = `

      INSERT INTO bookings (

        user_id, event_id, number_of_seats, total_price, status,

        ticket_holder_name, ticket_holder_email, ticket_holder_mobile,

        booking_date

      )

      VALUES ($1, $2, $3, $4, $5, $6, $7, $8, NOW())

      RETURNING *;

    `;
  }
}
```

```

const bookingValues = [
  user_id, event_id, number_of_seats, total_price, status || 'confirmed',
  ticket_holder_name, ticket_holder_email, ticket_holder_mobile];
const bookingResult = await client.query(bookingQuery, bookingValues);
const newBooking = bookingResult.rows[0];
// 2. Insert into payments table
const paymentQuery = `
  INSERT INTO payments (
    booking_id, amount, payment_method, transaction_id, status
  )
  VALUES ($1, $2, $3, $4, 'completed')
  RETURNING *;`;
// Use amount from body or fallback to total_price
const paymentAmount = amount || total_price;
await client.query(paymentQuery, [newBooking.id, paymentAmount, payment_method,
transaction_id]);
// 3. Update available seats
await client.query(
  'UPDATE events SET available_seats = available_seats - $1 WHERE id = $2',
  [number_of_seats, event_id]
);
await client.query('COMMIT');
console.log('✅ Booking created successfully:', newBooking.id);
res.status(201).json({
  message: 'Booking confirmed',
  booking: newBooking
});
} catch (err) {
  await client.query('ROLLBACK');
  console.error('❌ Database Error:', err);
}

```



```

    res.status(500).json({ error: 'Server error processing booking' });
  } finally {
    client.release();
  }
});
module.exports = router;

```

2.AUTH.JS

```

// Ensure services are loaded
if (typeof auth === 'undefined' || typeof Utils === 'undefined') {
  console.error('Services not loaded. Make sure config.js, utils.js, api.js, and auth.js are
included before this script.');
```

```

}
```

```

const form = document.getElementById('signup-form');
const passwordInput = document.getElementById('password');
const confirmPasswordInput = document.getElementById('confirm-password');
const errorMsg = document.getElementById('error-msg');
const successMsg = document.getElementById('success-msg');
```

```

// Back to login button listener
const backBtn = form.querySelector('button[type="button"]');
if (backBtn) {
  backBtn.addEventListener('click', () => {
    window.location.href = './login.html';
  });
}
```

```

// Submit signup form
form.addEventListener('submit', async (e) => {
  e.preventDefault();

```

```

// Clear previous messages
if (errorMsg) errorMsg.textContent = "";
if (successMsg) successMsg.textContent = "";

// Get form values
const name = document.getElementById('name').value.trim();
const mobile = document.getElementById('mobile').value.trim();
const role = document.getElementById('role').value;
const email = document.getElementById('email').value.trim();
const password = passwordInput.value;
const confirmPassword = confirmPasswordInput.value;

console.log('📝 Signup form submitted');
console.log('Form data:', { name, mobile, role, email, password_length: password?.length
});

// --- Validations ---
if (password !== confirmPassword) {
  if (errorMsg) errorMsg.textContent = 'Passwords do not match.';
  console.log('❌ Passwords do not match');
  return;
}

if (!Utils.validatePassword(password)) {
  if (errorMsg) errorMsg.textContent = 'Password must be at least 6 characters.';
  console.log('❌ Password too short');
  return;}

if (!Utils.validateEmail(email)) {
  if (errorMsg) errorMsg.textContent = 'Please enter a valid email address (e.g.,
user@example.com)';
  console.log('❌ Invalid email');
  return;}

```

```

if (!Utils.validateMobile(mobile)) {
  if (errorMsg) errorMsg.textContent = 'Please enter a valid 10-digit mobile number';
  console.log('❌ Invalid mobile');
  return;}

if (!Utils.validateName(name)) {
  if (errorMsg) errorMsg.textContent = 'Please enter a valid name.';
  console.log('❌ Invalid name');
  return;}

if (!role) {
  if (errorMsg) errorMsg.textContent = 'Please select a role.';
  console.log('❌ No role selected');
  return;}

console.log('✅ All validations passed');
// --- Disable button and show loading ---
const submitBtn = form.querySelector('button[type="submit"]');
submitBtn.disabled = true;
const originalText = submitBtn.textContent;
submitBtn.textContent = 'Creating Account...';
Utils.showLoading();
try {
  console.log('🔄 Sending signup request to backend...');
  const response = await auth.signup({
    name,
    mobile,
    role,
    email,
    password,
    confirmPassword
  });
}

```

```

console.log('✅ Signup response received:', response);

if (successMsg) successMsg.textContent = '✓ Account created successfully! Redirecting
to login...';

// Redirect after 1.5s
setTimeout(() => {
  window.location.href = './login.html';
}, 1500);

} catch (error) {
  console.error('❌ Signup failed:', error.message);
  console.error('Error details:', error);
  if (errorMsg) errorMsg.textContent = `Error: ${error.message}`;

  // Re-enable button and reset text
  submitBtn.disabled = false;
  submitBtn.textContent = originalText;

} finally {
  Utils.hideLoading();
}
});

```

APPENDIX II

SNAPSHOTS

8.1 Landing Page

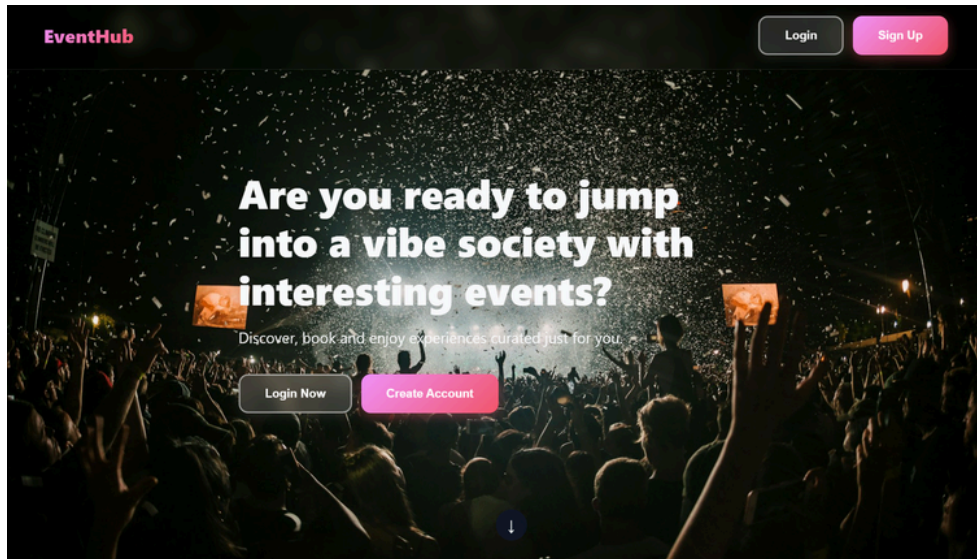


Fig 8.1 Home Page

8.2 About Page

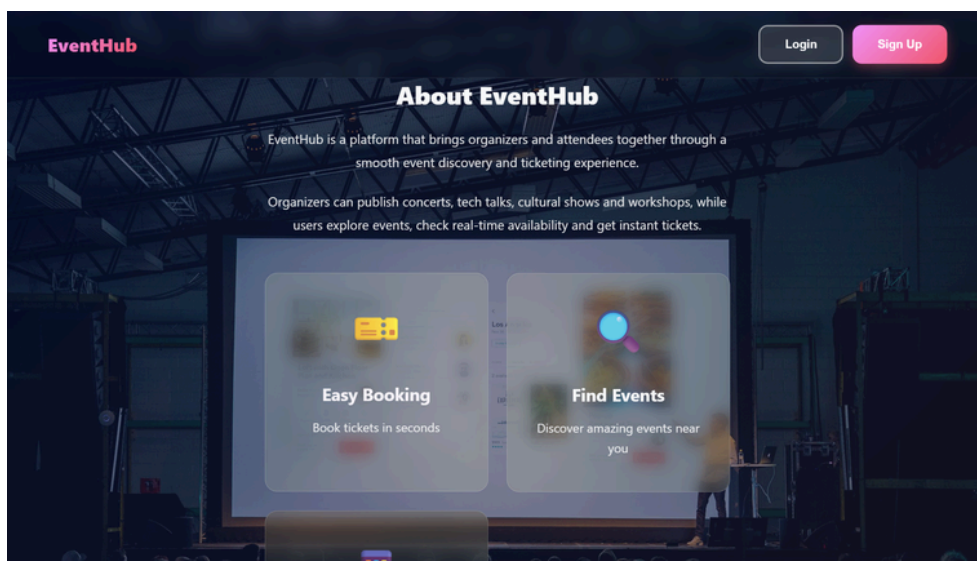


Fig 8.2 About Page

8.3 Contact Page

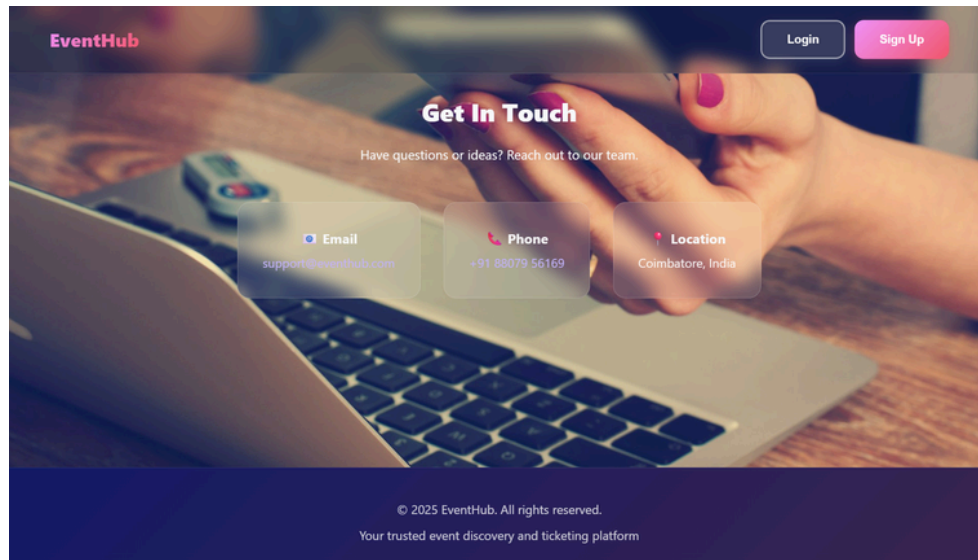


Fig 8.3 Contact page

8.4 Signup Page

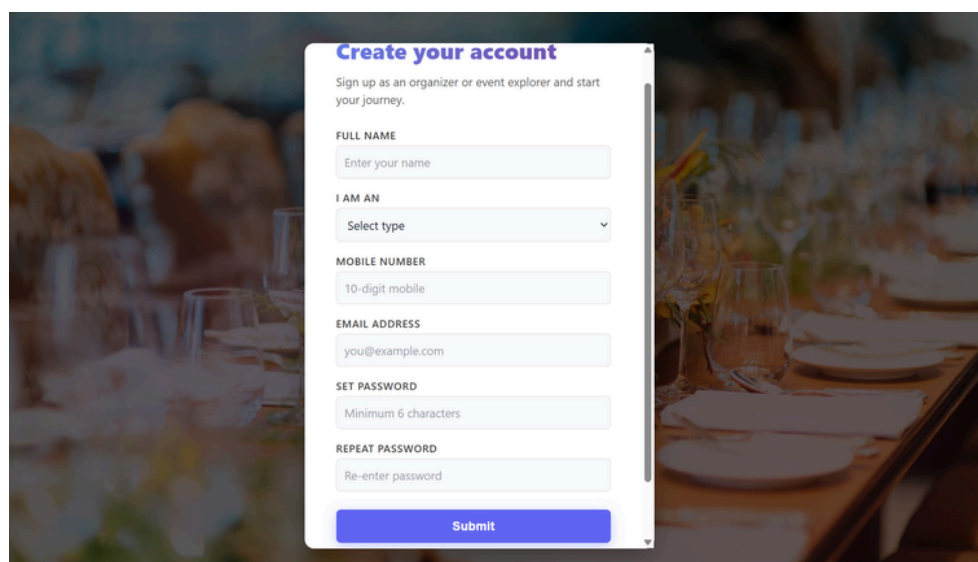
The image displays a 'Create your account' signup form overlaid on a background image of a restaurant table. The form includes the title 'Create your account' and the instruction 'Sign up as an organizer or event explorer and start your journey.' The fields are: 'FULL NAME' with a placeholder 'Enter your name'; 'I AM AN' with a dropdown menu 'Select type'; 'MOBILE NUMBER' with a placeholder '10-digit mobile'; 'EMAIL ADDRESS' with a placeholder 'you@example.com'; 'SET PASSWORD' with a placeholder 'Minimum 6 characters'; and 'REPEAT PASSWORD' with a placeholder 'Re-enter password'. A blue 'Submit' button is at the bottom of the form.

Fig 8.4 Signup Page

8.5 Login Page

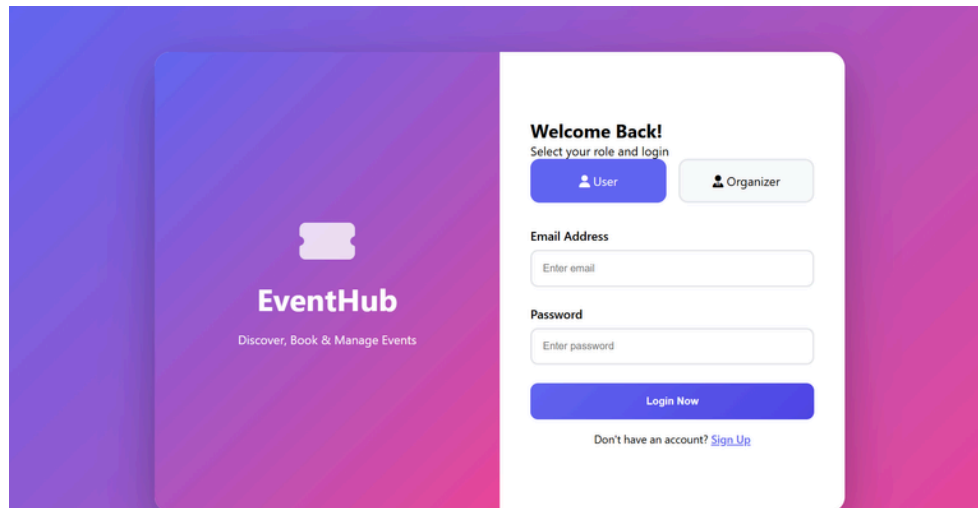


Fig 8.5 Login Page

8.6 Admin Dashboard Page

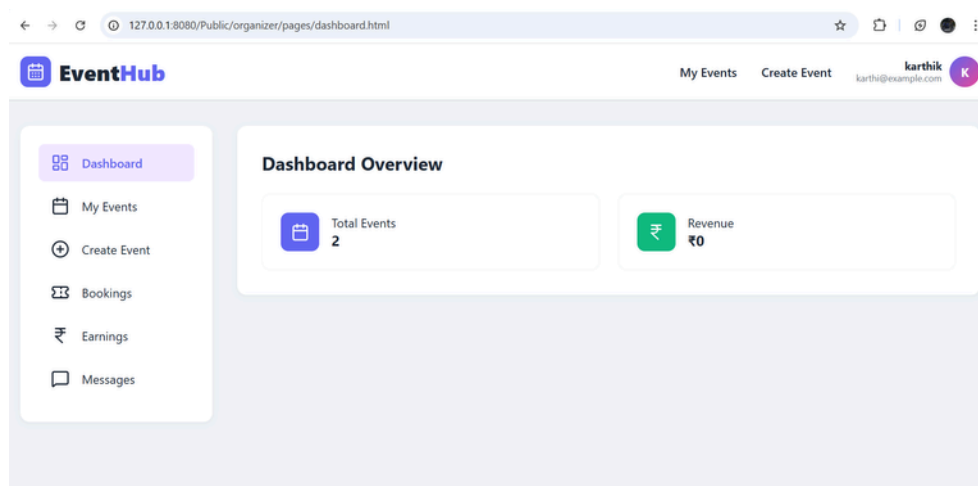


Fig 8.6 Admin Dashboard

8.7 Admin Events Page

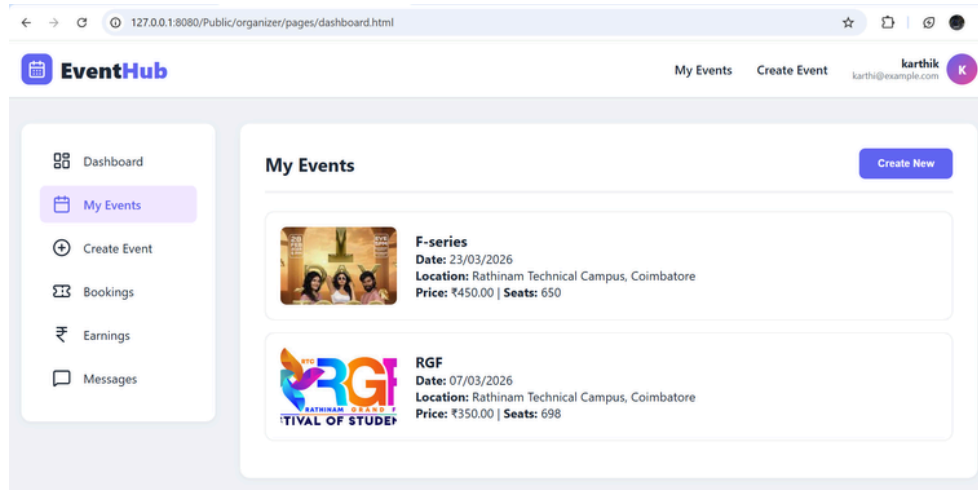


Fig 8.7 Admin Events Page

8.8 Admin Profile Page

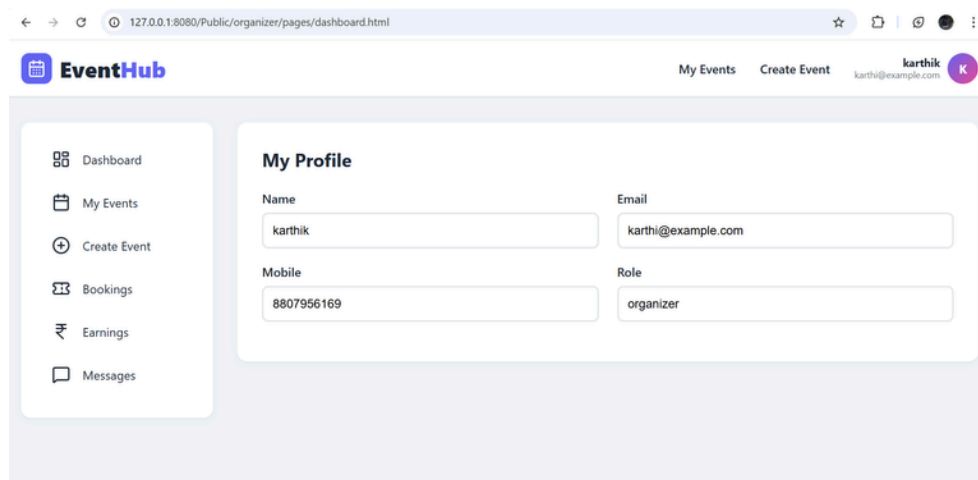


Fig 8.8 Admin Profile Page

8.9 Event Creation Page

EventHub

My Events Create Event karthik K karthi@example.com

Create New Event

Event Title

Category

Date

Location

Price (₹)

Seats

Description

Upload Images No file chosen

Fig 8.9 Event Creation Page

8.10 User Dashboard Page

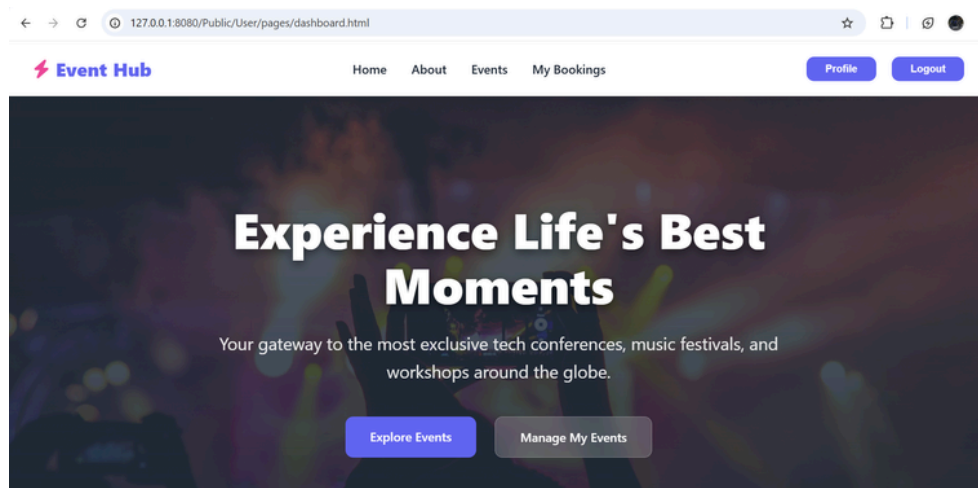


Fig 8.10 User Dashboard Page

8.11 User Events Page

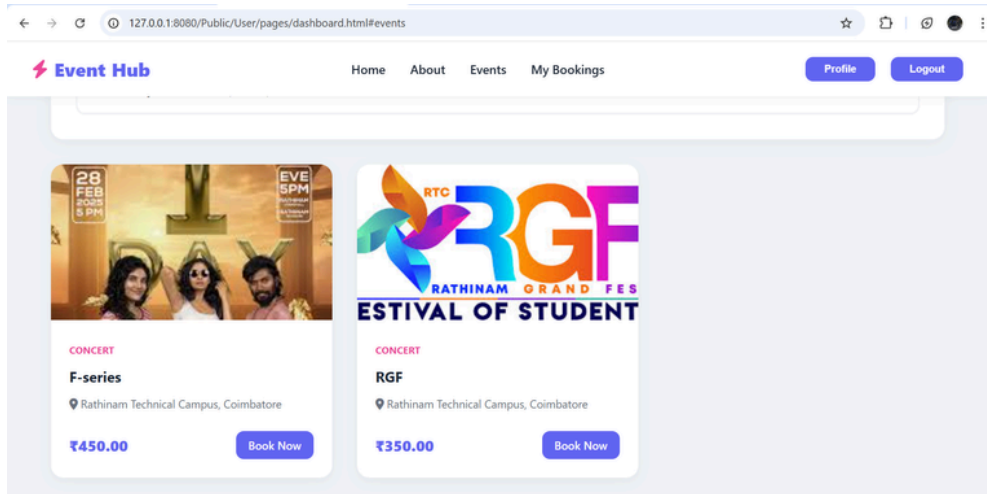


Fig 8.11 User Events Page

8.12 User Booking History Page

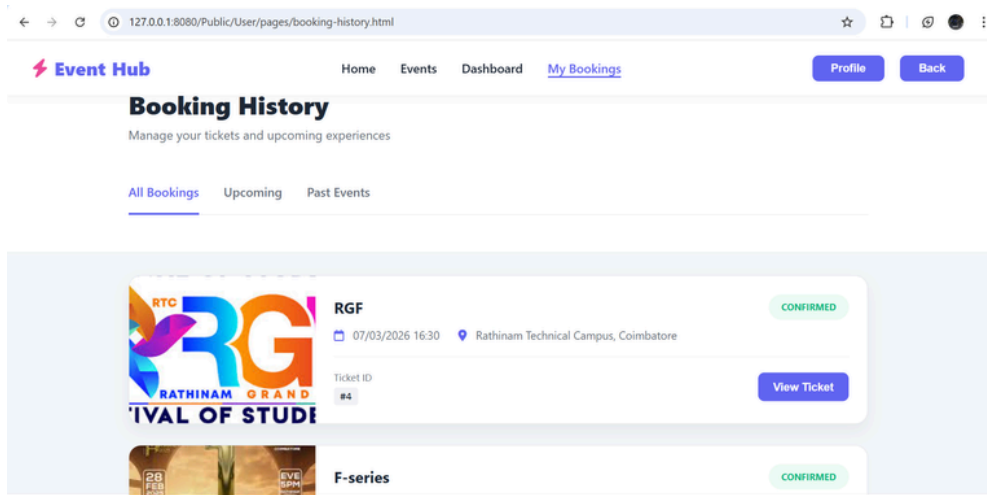


Fig 8.12 User Booking History Page

8.13 User Profile Page

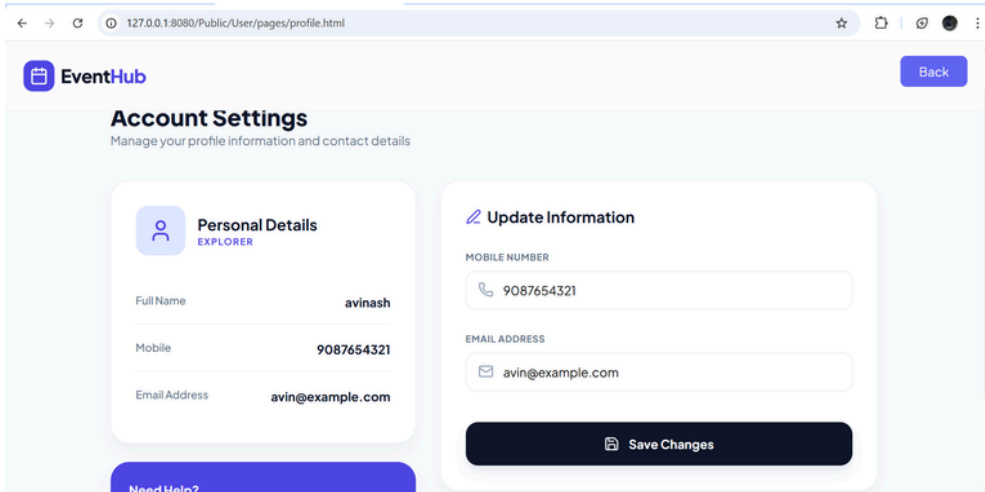


Fig 8.13 User Profile Page

8.14 Event Details Page

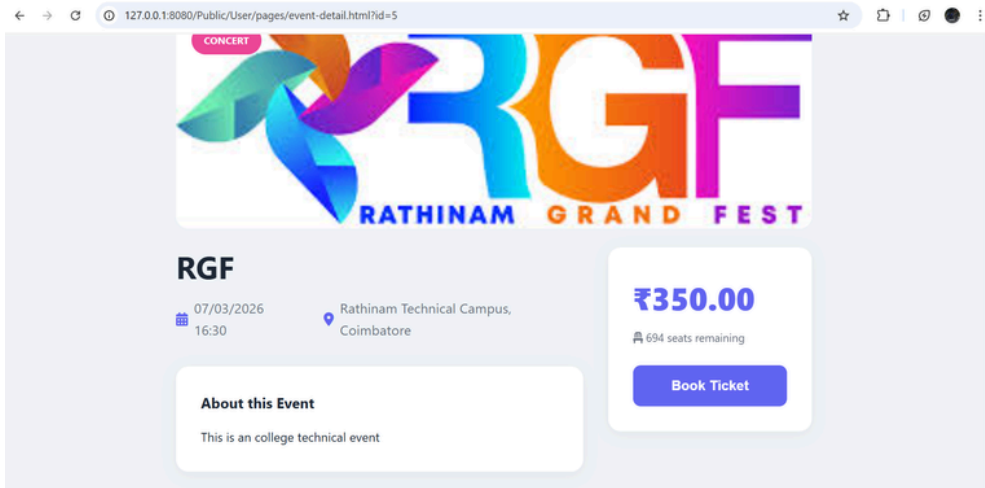


Fig 8.14 Event Details Page

8.15 Booking Page

The screenshot shows a web browser window with the URL `127.0.0.1:8080/Public/User/pages/booking.html?id=5`. The page is titled "EventHub" and features a "Back" button in the top right corner. The main content area is divided into three sections:

- Event Details:** A purple box labeled "SELECTED EVENT" for "RGF" on "01/05/2024, 10:30:00" at "Railways Technical Campus, Coimbatore". The price is "₹500.00 per person".
- Order Summary:** A white box showing "Tickets: 4" for "₹1400", "Service Fee: ₹0.00", and a "Total Amount: ₹1400".
- Attendee Information:** A white box with two steps: "1. Attendee Information" and "2. Payment Method".
 - Attendee Information:** Includes "Number of Tickets" (4), "Ticket Holder Name" (anishah), "Card Number" (4242 4242 4242 4242), "expiry" (12/30), "CVC" (***), "Email Address" (anishah@example.com), and "Phone Number" (9087654321).
 - Payment Method:** Includes "Card" and "UPI" options.

A "Confirm & Pay ₹ 1400" button is located at the bottom right of the Attendee Information section.

Fig 8.15 Booking Page

8.16 Booking Confirmation Page

The screenshot shows a "Booking Confirmed!" message with a green checkmark icon. Below the message, it states "Your tickets have been successfully booked". The page is divided into two main sections:

- Ticket Details:** A list of information including "Ticket ID: T01-AMRQJPM", "Event: RGF", "Ticket Holder: anishah", "Number of Tickets: 4", "Event Date/Time: 01/05/2024, 10:30:00", "Location: Railways Technical Campus, Coimbatore", "Total Amount: ₹1400", "Payment Method: Credit Card", and "Transaction ID: T24-162199C34".
- Your Ticket QR Code:** A section with the text "Show this QR code at the event entrance" and a QR code placeholder.

At the bottom, there are two buttons: "Download Ticket" and "Back to Home".

Fig 8.16 Booking Confirmation Page